Linear Algebra - Section 1.1
Systems of Linear Equations

- Def: A linear equation w/ the variables $x_1, x_2, \dots, x_n$ is an equation that can be writen in the form:

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = b$$

where a_i, $1 \le i \le n$, and b are real (or complex) numbers called the coefficients.

- Def: A system of linear equations is a collection of one OR more linear equations.
  → There are 3 solution types:
  1. One solution (a point/ordered pair) $(x_1, x_2, \dots, x_n)$
  2. Infinitely many solutions (for 2D systems, the solution is a line; for 3D systems the solution could be a line OR plane).
  3. No solution

  → There ar 2 types of systems:
  1. Consistent: there is a solution! (one OR many).
  2. Inconsistent: No solution

- Examples:
  A 2D system: $\begin{cases} 2x - y = 3 \\ x + 4y = 5 \end{cases}$

  2 equations,
  2 unknowns,
  → lives in the 2D plane (xy-coordinates)

A 3D system: $\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ \quad\quad x_2 - 7x_3 = 12 \\ 5x_1 - x_2 + 6x_3 = 4 \end{cases}$

3 equations,
3 unknowns
→ lives in the 3D coordinate system (x, y, z)

- We can compactly write/record the information of a linear system in a rectangular array called a **matrix**:

$$\begin{array}{c} \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} \\ \uparrow \quad \uparrow \\ x \quad\ y \end{array} \quad \text{OR} \quad \left[\begin{array}{cc|c} 2 & -1 & 3 \\ 1 & 4 & 5 \end{array}\right]$$

* called the **coefficient matrix**

* called the **augmented matrix**

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -7 & 12 \\ 5 & -1 & 6 & 4 \end{array}\right]$$ ← augmented matrix for the previous 3D system

* Note that the bar may or may not be there (your book!) I like the bar, so I'll usually include it.

- The size of a matrix tells us how many rows and columns it has:

$$\underset{\substack{\uparrow \\ \#\text{ of} \\ \text{Rows}}}{m} \times \underset{\substack{\nwarrow \\ \#\text{ of} \\ \text{columns}}}{n}$$

- GOAL: Learn how to use a matrix to solve a linear system!

- Ex: $\begin{cases} 2x_1 + 8x_2 + 4x_3 = 2 \\ 2x_1 + 5x_2 + x_3 = 5 \\ 4x_1 + 10x_2 - x_3 = 1 \end{cases}$

Step 1: Convert to an augmented matrix:

$$\left[\begin{array}{ccc|c} 2 & 8 & 4 & 2 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array}\right]$$

Step 2: Solve! (umm... ok? But how?!)

The How: Elementary Row Operations

1. Replacement: Replace a row by the sum of itself and a multiple of another row.
2. Interchange: Interchange (aka swap) two rows.
3. Scaling: multiply all entries in a row by a nonzero constant.

* Row operations are reversible!
* Two matrices are called row equivalent if there is a sequence of elementary row operations that transform one matrix into another.
* Row equivalent matrices have the same solution set!

Now, back to the example:

$$\begin{array}{c} R_1 \rightarrow \\ R_2 \rightarrow \\ R_3 \rightarrow \end{array} \left[\begin{array}{ccc|c} 2 & 8 & 4 & 2 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array}\right]$$

I'm called a pivot.

$$\tfrac{1}{2}R_1 \rightarrow \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array}\right]$$

$$-2R_1 + R_2 \rightarrow R_2: \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & -3 & -3 & 3 \\ 4 & 10 & -1 & 1 \end{array}\right]$$

this Row changes

$$-4R_1 + R_3 \rightarrow R_3: \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & -3 & -3 & 3 \\ 0 & -6 & -9 & -3 \end{array}\right]$$

$$-\tfrac{1}{3}R_2 \rightarrow \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & -6 & -9 & -3 \end{array}\right]$$

New pivot!

$$6R_2 + R_3 \rightarrow R_3: \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -3 & -9 \end{array}\right]$$

$$-\tfrac{1}{3}R_4 \rightarrow \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

* I can stop now and easily solve the system!

Convert the matrix back to equation form:

$$\begin{cases} x_1 + 4x_2 + 2x_3 = 1 \\ \quad\quad\; x_2 + x_3 = -1 \\ \quad\quad\quad\quad \boxed{x_3 = 3} \end{cases}$$

Nice!

$x_3 = 3$

Plug into 2nd equation

$x_2 + x_3 = -1$

$x_2 + (3) = -1$

$x_2 = -4$

Now plug both into 1st equation

$x_1 + 4x_2 + 2x_3 = 1$

$x_1 + 4(-4) + 2(3) = 1$

$x_1 - 16 + 6 = 1$

$x_1 - 10 = 1$

$x_1 = 11$

Solution: $(11, -4, 3)$

* OUR system is consistent w/ one unique solution!
* Graph it!

Linear Algebra - Section 1.2
Row Reduction and Echelon Forms

- Def: A rectangular matrix is in echelon form (aka row echelon form) if it has the following 3 properties:

    1. Any nonzero rows are above any rows of all zeros.
    2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
    3. All entries in a column below a leading entry are zeros.

If a matrix in echelon form also satisfies the following 2 additional properties, then it is in reduced echelon form (aka reduced row echelon form):

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.

basically all 0's and 1's

- We can tell if a system is consistent OR inconsistent by converting it to an augmented matrix and performing row operations to transform it into echelon OR reduced echelon form.

* check to see if there is any row of the form:

  $[0\ 0\ 0 \cdots 0 \mid b]$

* If $b \neq 0$, then the system is inconsistent and no solution exists.
* If $b = 0$, then the system is consistent w/ infinitely many solutions.
* The number of all zero rows tells us how many free variables we have.

unassigned (could be anything)